Tab 1

Student Handout

Name: _Gabrielle D. Gabrielle S. Ethan R, Tez___ **Date:** _____ **Class:** _____

MyDesign Portfolio

Element D Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your design ideas, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own design ideas, be sure to delete the instructions.

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

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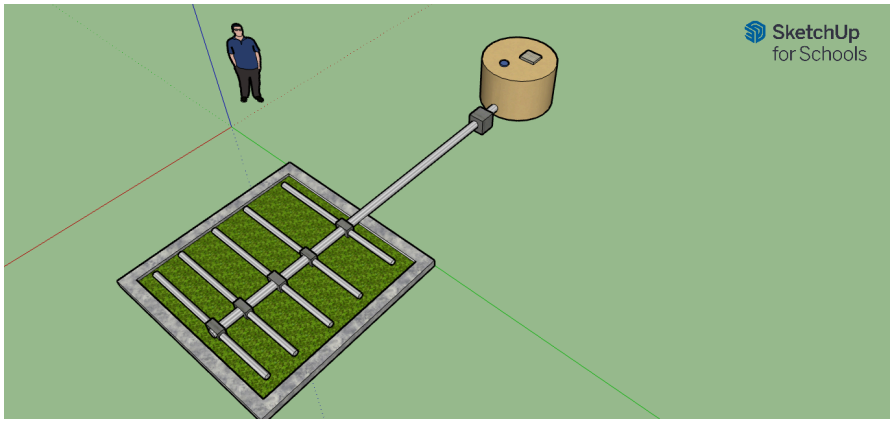
Element D: Design Concept Generation, Analysis, and Selection

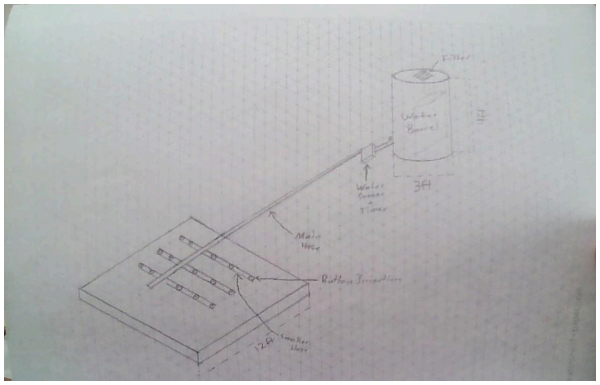
Design Idea Generation *(D1)*

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Design Idea #1



Date: 2/26/25
Description: Irrigation system using PVC piping ½ in (10ft rows), PVC fittings 3/4in (T and 4-way), garden hose, a water holding tank, timer/regulator, and filter for debris from roof runoff water.
Drawing that includes distinguishing information such as scale, dimensions, and materials:



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Design Idea Comparison and Selection *(D1, D2)*

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

Ethan Roadfuss	Weight as decimal	Pressure Gauge	Water Barrel	Timer	Filter/Grate	Adapter	Sprinkler/Water dropper
Irregate a 12x12 area	0.3	1	0	1	1	1	0
Last 10 days	0.2	1	0	0	0	0	0
Use collected water	0.1	1	1	0	1	1	0
Portable	0.1	0	-1	0	0	0	0
Above ground	0.1	0	1	0	0	0	0
Sustainable/Efficient	0.2	1	1	1	1	1	0
	total score	0.8	0.3	0.5	0.6	0.6	0

Weight Reasoning: We chose to weigh each criteria based on the stockholders' needs. The #1 priority is to irrigate the 12x12 area, hence making it the highest weight. It is important to the stockholder that it lasts 10 days and that the system is efficient. Lastly, using collected water, being portable, and being above ground are all requested criteria but are not necessarily required to manufacture an adequate solution.

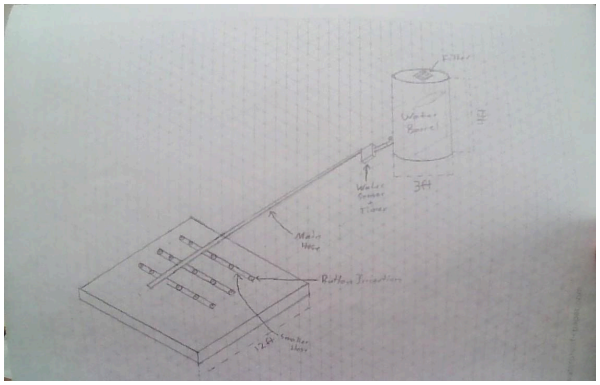
Design Blueprint *(D4)*

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

Name: _Gabrielle D. Gabrielle S. Ethan R, Tez___ **Date:** _____ **Class:** _____

Date: 2/26/25
Sketch:



Manufacturing Methods:

Step 1: Order and gather materials.

Step 2: Layout and measure PVC, cutting in parts that need to be shortened or placed into a fitting.

Step 3: Put together all PVC into fittings and create as planned in blueprint and sketch and connect to the water source.

Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
PVC Pipe (10ft) (1/2in) link	$4.71	6	$28.26
T-Fitting (3/4in) link	$0.82	1	$0.82
4-way Fittings (3/4in) link	$5.21	4	$20.84
PVC to garden hose adapter (3/4in)	$2.98	1	$2.98

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link			
PVC end caps 3/4in link	$0.82	10	$8.20
Regulator/timer link	$19.94	1	$19.94
Total Cost	$32.17	23	$78.73

Plan of Action *(D3)*

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

1. Create a design and sketch of intending product making sure to follow all of the guidelines.
2. Research materials to be used in the product.
3. Order and gather materials needed.
4. Lay out materials making sure everything will work as intended.
5. Measure and cut pipes of PVC to fit the space of the garden and the design.
6. Drill holes to let water drip to the plants and add caps to the ends of the PVC.
7. Connect the pipe to the fittings to create the products grid like design.
8. Connect the timer/regulator to the initial water source.
9. Connect hose from the timer to the PVC female adaptor.
10. Then to the system of the PVC watering.
11. Test - revise - approve

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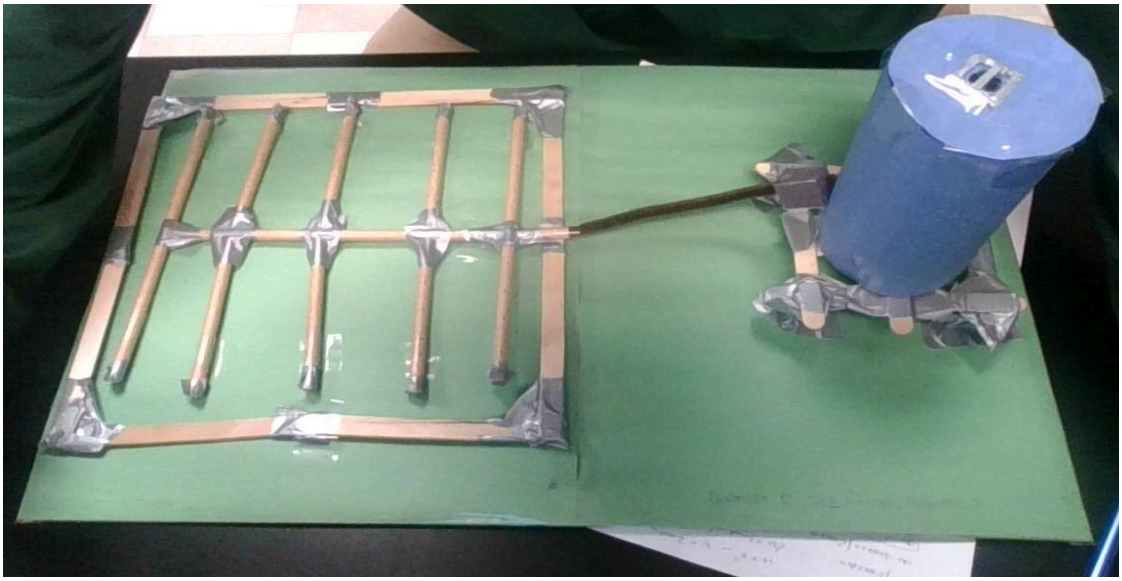
Model